



SECI1113: COMPUTATIONAL MATHEMATICS
20232024 – SEMESTER 2

ASSIGNMENT #1

(Deadline of Submission: 16th Apr. 2024; 5 pm)

1. A construction company is building three types of houses: small, medium, and large. Each house requires different quantities of bricks (b) in 100 units, cement (c) in 100 bags, and wood (w) in 100 cubic feet. The company's construction process can be represented by the following system of linear equations:

For the small house: $13b + 8c + 5w$

For the medium house: $15b + 12c + 10w$

For the large house: $20b + 17c + 15w$

- a) Write the system of linear equations in matrix form and label it as matrix A. (2 marks)
- b) Find the inverse of matrix A using the formula: $A^{-1} = \frac{1}{\det(A)} \text{adj}(A)$. (8 marks)
2. The company's production process can be represented by the following system of linear equations:
- $$2p + 3m + b = 5$$
- $$3p + 4m + 2b = 8$$
- $$p + 2m + 3b = 10$$
- Solve this system of linear equations using **Gaussian elimination**. (10 marks)
3. Given matrices P and Q, determine which of the matrices has an inverse. Then, for those that have an inverse, find the inverse using a linear system. (15 marks)

$$P = \begin{bmatrix} 1 & 2 & 1 \\ 2 & 4 & -1 \\ 3 & 6 & 0 \end{bmatrix}; \quad Q = \begin{bmatrix} 2 & 7 & 4 \\ 5 & 19 & 11 \\ 3 & 13 & 8 \end{bmatrix}$$

4. Perform *LU* factorization using (i) **Doolittle** and (ii) **Crout** methods to solve the following system of linear equations.

$$A = \begin{bmatrix} 4 & 2 & 3 \\ 3 & 3 & 2 \\ 2 & 4 & 2 \end{bmatrix} \quad B = \begin{bmatrix} 18 \\ 16 \\ 17 \end{bmatrix}$$

(20 marks)

5. Let, $u = (5, 4, -3)$ and $v = (1, -2, -1)$. Find,

- a) The Euclidean distance between u and v .
- b) The dot product, with $\theta = 45^\circ$
- c) The cross product.

(5 marks)

6. Let,

$$v_1 = \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}; v_2 = \begin{bmatrix} 2 \\ 1 \\ 1 \end{bmatrix}; v_3 = \begin{bmatrix} 0 \\ 2 \\ 1 \end{bmatrix} \text{ and } v_4 = \begin{bmatrix} -2 \\ 2 \\ 1 \end{bmatrix}$$

Show that v_4 can be written as a linear combination of v_1, v_2 and v_3 .

(5 marks)

7. Consider the vectors, $u = (1, -3, 2)$ and $v = (2, -1, 1)$ in R^3 .

- a) For which value of k is $(1, k, 5)$ is a linear combination of u and v ?
- b) Determine whether u and v are linearly independent.

(5 marks)